

TASK WK1-PM1

Cybersecurity Testing Lab Environment Setup

VirtualBox + Kali Linux + Custom NAT Network (10.0.0.0/24)

Report Details	
Course	Cybersecurity & Ethical Hacking – Networkwalks Academy
Assignment	WK1-PM1: Setup a Cybersecurity Testing Lab Environment
Environment	Oracle VirtualBox with Kali Linux VM
Network Mode	Custom NAT Network — “NatNetwork” (10.0.0.0/24)
Kali Linux IP	10.0.0.2 /24, Gateway 10.0.0.1, DNS 8.8.8.8

1. Objective

The goal of this task was to build a cybersecurity testing lab on a personal laptop/PC using Oracle VirtualBox, with Kali Linux configured as the attacking/hacker machine. The lab network had to be isolated on the 10.0.0.0/24 subnet using a custom NAT Network, with Kali Linux assigned a static IP of 10.0.0.2/24 and full internet access, in line with the specification below.

Task Requirements (as issued)

- Use VirtualBox as the base hypervisor (latest recommended version).
- Set up Kali Linux as the attacking/hacker machine.
- Set up the lab network in subnet 10.0.0.0/24.
- Use a NAT Network named to match, configured for 10.0.0.0/24.
- Enable clipboard sharing and file drag-and-drop in the VM settings.
- Enable shared folders, with the host machine's /downloads folder shared.
- Assign Kali Linux the static IP address 10.0.0.2/24.
- Confirm Kali Linux has full internet access.

2. Lab Architecture

The lab follows the standard Networkwalks architecture: the host machine (main OS) runs Oracle VirtualBox, inside which a custom NAT Network called “NatNetwork” provides an isolated 10.0.0.0/24 subnet for all lab virtual machines. Kali Linux acts as the attacking machine on this subnet and reaches the internet through the NAT Network's built-in router/DHCP service, without exposing the lab to the host's main LAN.

Component	Configuration
Hypervisor	Oracle VirtualBox (latest version)
Attack VM	Kali Linux
Network Type	NAT Network (custom, not default NAT or Bridged)

Network Name	NatNetwork
Subnet / IPv4 Prefix	10.0.0.0/24
DHCP	Enabled on the NAT Network
Kali IP Addressing	Manual/Static — 10.0.0.2, Netmask 24, Gateway 10.0.0.1
DNS	8.8.8.8 (fallback: 10.0.0.1 if resolution issues occur)

3. Implementation Steps & Evidence

3.1 Creating the Custom NAT Network

In VirtualBox Manager, a NAT Network named “NatNetwork” was created via File → Tools → Network Manager, using IPv4 Prefix 10.0.0.0/24 with DHCP enabled. This gives every VM attached to it an isolated internal subnet that can still reach the internet.

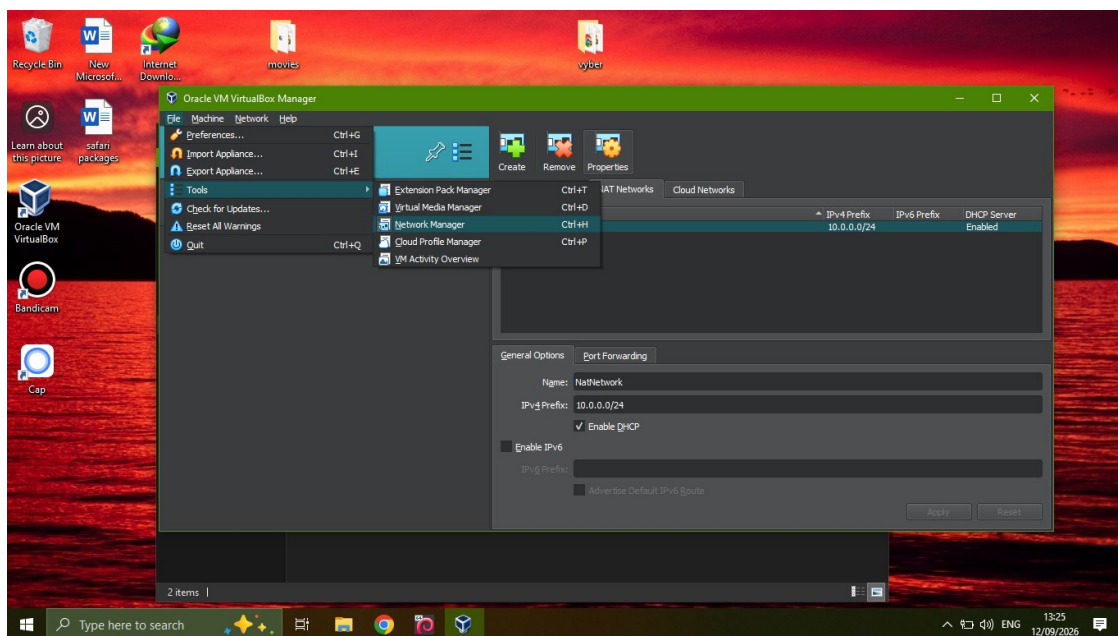


Figure 1 – VirtualBox NAT Networks manager showing “NatNetwork” with IPv4 Prefix 10.0.0.0/24 and DHCP Server Enabled.

3.2 Attaching the Kali Linux VM to the NAT Network

The Kali Linux VM's Network settings (Adapter 1) were configured to attach to the custom NAT Network rather than the default NAT adapter, with the cable marked as connected so the interface comes up automatically at boot.

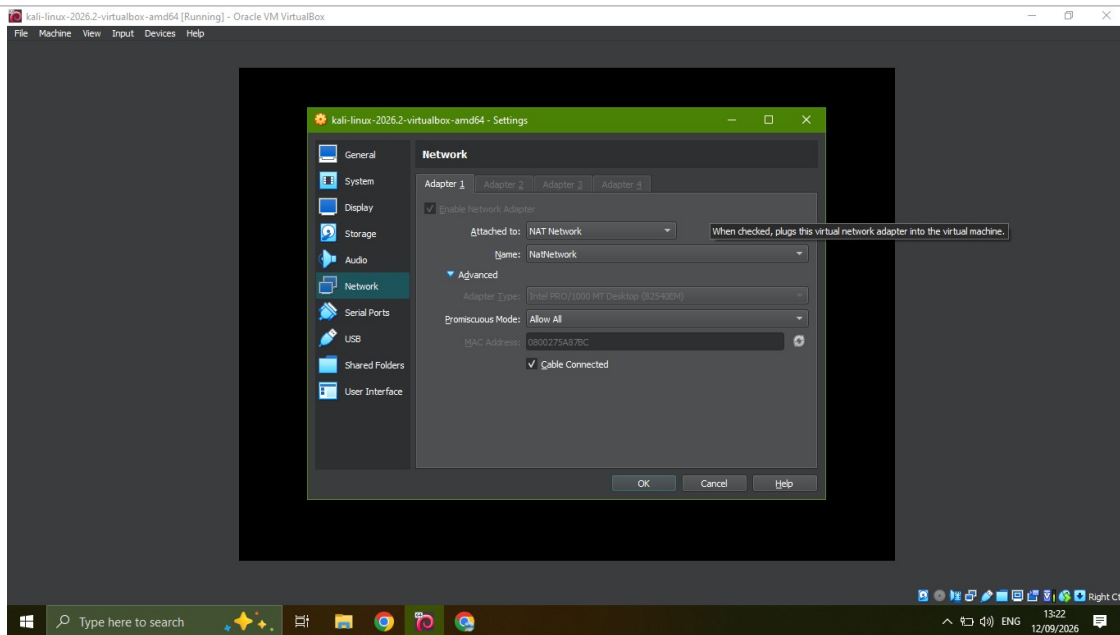


Figure 2 – Kali Linux VM – Adapter 1 attached to “NAT Network”, Name: NatNetwork, Cable Connected.

3.3 Configuring the Static IP on Kali Linux

Inside Kali Linux, the wired connection's IPv4 settings were switched from Automatic (DHCP) to Manual, and the address required by the task — 10.0.0.2/24 with gateway 10.0.0.1 — was applied, using 8.8.8.8 as the DNS server.

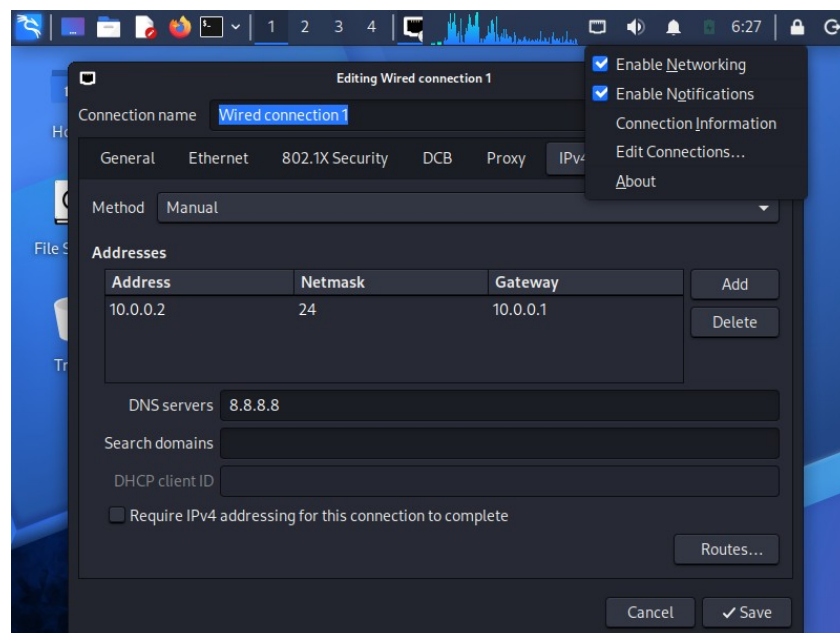


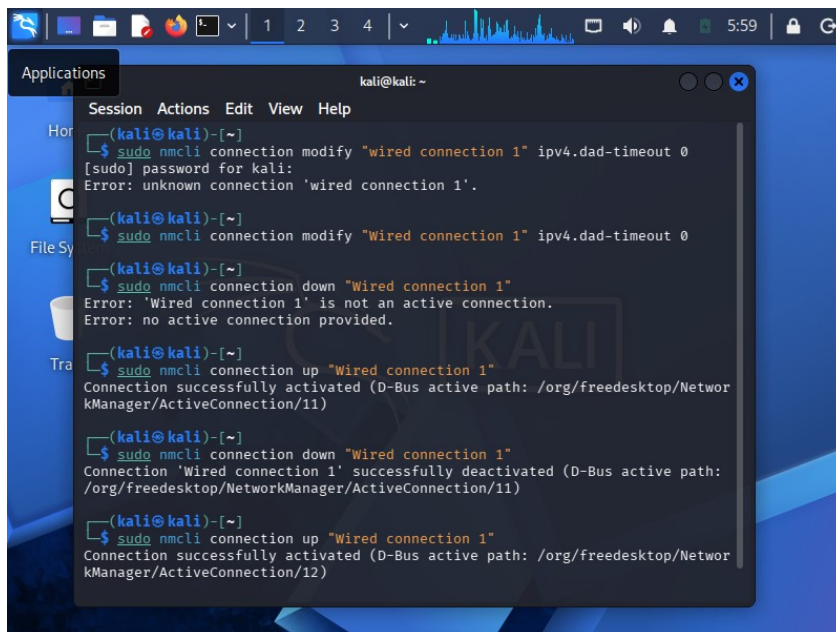
Figure 3 – Kali Linux “Wired connection 1” IPv4 settings: Address 10.0.0.2, Netmask 24, Gateway 10.0.0.1, DNS 8.8.8.8.

3.4 Troubleshooting Internet Connectivity

After applying the static IP, Kali Linux initially did not have working internet access — a known issue on VirtualBox v7 with recent Kali Linux releases, caused by IPv4 duplicate-address-detection (DAD) delaying the interface. This was resolved from the terminal using the recommended nmcli commands to disable the DAD timeout and cycle the connection:

```
sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0
sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```

The terminal output confirmed the connection was successfully deactivated and then reactivated ("Connection successfully activated - D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/12"), after which Kali Linux obtained full internet access.



```
kali@kali: ~
$ sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0
[sudo] password for kali:
Error: unknown connection 'Wired connection 1'.

(kali@kali)-[~]
$ sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0
Error: 'Wired connection 1' is not an active connection.
Error: no active connection provided.

(kali@kali)-[~]
$ sudo nmcli connection up "Wired connection 1"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/11)

(kali@kali)-[~]
$ sudo nmcli connection down "Wired connection 1"
Connection 'Wired connection 1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/11)

(kali@kali)-[~]
$ sudo nmcli connection up "Wired connection 1"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/12)
```

Figure 4 – Terminal output resolving the connectivity issue and confirming the connection was brought back up successfully.

3.5 Enabling Shared Clipboard & Drag-and-Drop

In the Kali Linux VM's General → Advanced settings, Shared Clipboard and Drag'n'Drop were both set to Bidirectional, allowing copy/paste and file drag-and-drop to work in both directions between the host machine and the Kali VM.

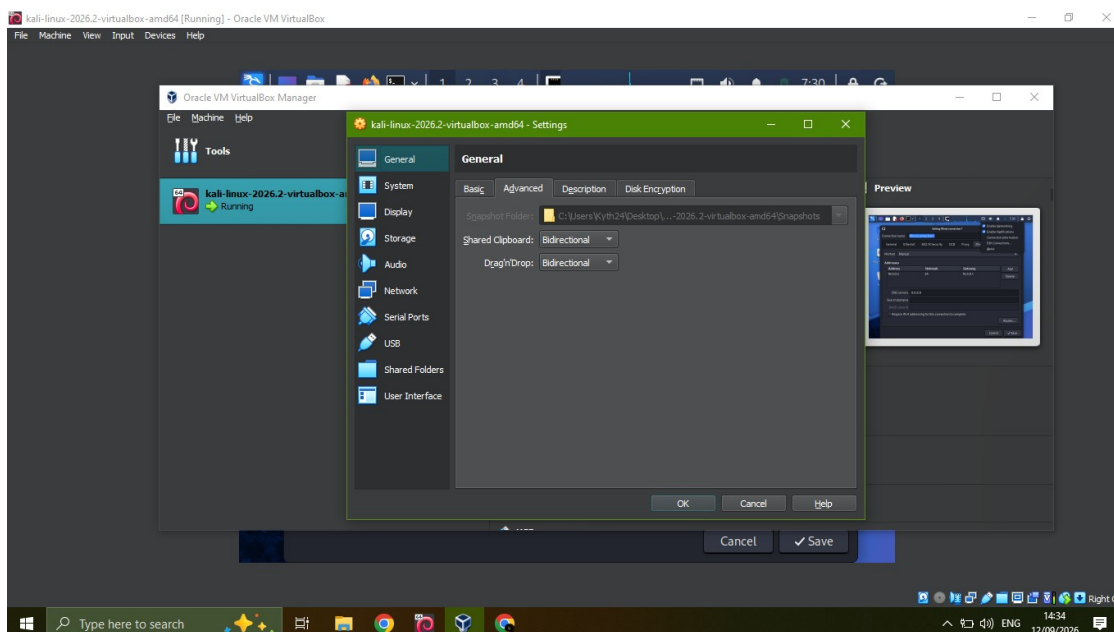


Figure 5 – Kali Linux VM – General → Advanced: Shared Clipboard and Drag'n'Drop both set to Bidirectional.

3.6 Enabling the Shared Folder

A shared folder was added under Settings → Shared Folders, mapping a folder on the host machine to the Kali VM with Full access and Auto-mount enabled. The host path used is C:\Users\Kyth24\Downloads\Cyber Intern — a subfolder inside the host's Downloads directory.

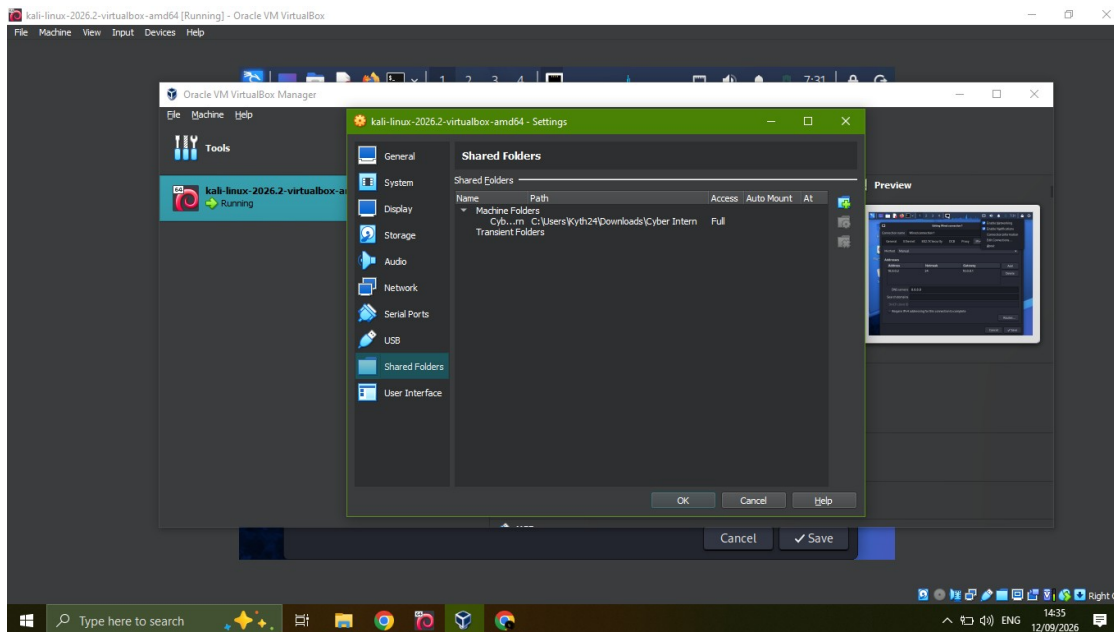


Figure 6 – Kali Linux VM – Shared Folders: host folder mapped with Full access and Auto Mount enabled.

Note: the task asked for the host's /downloads folder itself to be shared. The folder currently mapped is a subfolder named "Cyber Intern" inside Downloads, rather than the Downloads folder root. Functionally this still gives Kali access to shared files, but if the requirement is graded literally, either share the Downloads folder itself instead, or rename/move this mapping so the shared root is Downloads.

4. Requirements Checklist

Status of each item from the original task brief, based on the configuration evidenced above:

- ☒ VirtualBox installed and used as the base hypervisor.
- ☒ Kali Linux set up as the attacking/hacker machine.
- ☒ Lab network created in subnet 10.0.0.0/24.
- ☒ Custom NAT Network ("NatNetwork") used for 10.0.0.0/24.
- ☒ Kali Linux IP address set to 10.0.0.2/24.
- ☒ Kali Linux confirmed to have full internet access.
- ☒ Clipboard sharing & drag-and-drop enabled in VM settings (Bidirectional).
- ☒ Shared folder enabled from host machine, mounted with Full access.

One caveat: the shared folder currently maps a subfolder ("Cyber Intern") inside the host's Downloads directory rather than the Downloads folder itself — see section 3.6 for detail and a suggested fix if the requirement needs the literal Downloads root shared.

5. Conclusion

The cybersecurity testing lab was successfully built on VirtualBox using a custom NAT Network (NatNetwork) scoped to 10.0.0.0/24, with Kali Linux attached as the attacking machine and manually addressed at 10.0.0.2/24. An initial internet connectivity issue caused by IPv4 DAD timeout on newer Kali/VirtualBox v7 builds was diagnosed and resolved using nmcli, after which the Kali VM obtained full internet access. Bidirectional shared clipboard and drag-and-drop were enabled, and a host folder was shared into the VM with full access. All eight task requirements are now evidenced, with one minor note on the shared folder path (see 3.6). This lab environment is now ready to be extended in Phase 2 with additional victim machines (Windows, Server, Android) on the same NAT Network for future ethical hacking practice.